

Math Camp

Lecture 1

Dirghayu Shah

Language

- The language for real numbers will consist of constants, variables, grammatical symbols, connectives, quantifiers, functions and relations.
 - for all ($\forall$) and there exists ($\exists$) are quantifiers.
 - and ($\wedge$), or ($\vee$), not ($\neg$), implies ($\Rightarrow$) and iff ($\Leftrightarrow$) are connectives
- A statement in a language is said to be a sentence if every variable appearing in it is bound by a quantifier.
- Any sentence is either true or false, but not both.

Language

- A **connective** is an operator that combines or modifies one or more logical variables or formulas
- Negation ($\neg$)
 - $\neg a$ is false when a is true.
- Conjunction ($\wedge$, *and*)
 - $a \wedge b$ is true, *if and only if* (iff) both a and b are true.
- Disjunction ($\vee$, *or*)
 - $a \vee b$ is true, iff either a is true or b is true.
 - Note that *or* is inclusive, both a and b might also be true.

Language

- $a \wedge b$ is logically equivalent to $\neg(\neg a \vee \neg b)$

Language

- $a \wedge (b \vee c)$ is logically equivalent to $(a \wedge b) \vee (a \wedge c)$

Language

- Conditional ($\Rightarrow$)
 - $a \Rightarrow b$ means that if a is true, then b is true, or b if a .
 - Here a is the *antecedent* and b is the *consequent*
 - When $a \Rightarrow b$ holds, this implies that
 1. b is a **necessary condition** for a
 - a cannot be true without b being true
 2. a is a **sufficient condition** for b
 - a being true guarantees b being true
- You need to be at least 35 years old to become the President of the United States
 - Being at least 35 years old does not make you the President

Language

- $a \Rightarrow b$ and $\neg a \vee b$ are logically equivalent

Language

- Biconditional ($\Leftrightarrow$)
 - $a \Leftrightarrow b$ or a iff b means that either a and b are both true or both false.
 - a is a **necessary and sufficient condition** for b
 - b is a **necessary and sufficient condition** for a
 - a and b are *logically equivalent*
- “Jake is an unmarried man” and “Jake is a bachelor” are logically equivalent statements.

Language

- $a \Leftrightarrow b$ is logically equivalent to $(a \Rightarrow b) \wedge (b \Rightarrow a)$

Language

- The symbols $\forall$ (*for all* or *for every*) and $\exists$ (*there exists*) are **quantifiers**.
- A sentence is well formulated only if every variable is bound by a quantifier.
 - If a variable is not bound by a quantifier then the sentence is ambiguous, and may be neither true nor false.
 - “Cats are black” ✗
 - There are cats that are black, but not every cat is black
 - “Some cats are black” ✓

Language

Let $\mathcal{S}$ denote some statement about x

- $\forall x \mathcal{S}$ is logically equivalent to $\neg \exists x \neg \mathcal{S}$

- $\exists x \mathcal{S}$ is logically equivalent to $\neg \forall x \neg \mathcal{S}$

Language

- The order of quantifiers matter
- Let $\mathcal{S}$ denote some statement about x and y .
 $\forall y \exists x \mathcal{S}$ and $\exists x \forall y \mathcal{S}$ are two distinct claims
 - ▶ In the **first claim**, each y has a potentially different x .
 - ▶ In the **second claim**, the same x exists for all y .
- *Everybody has a mother* is not the same as
A mother for everybody.
- *For every mouse there is a piece of cheese* is not the same as
There is a piece of cheese for every mouse.
 - ▶ The two sentences are understood to mean the same in natural language, but not in mathematics

Proofs

Direct or Deductive Proofs

- You use a sequence of deductions, to ultimately establish the truth.
- These are the simplest kinds of proofs.
- **Example:** Prove that $\forall x, y, z \in \mathbb{R}$, if $x + z = y + z$ then $x = y$.
 1. We are given that, $x + z = y + z$
 2. We know that, $z - z = 0$
 3. When we add $-z$ to (1), we get $x + z - z = y + z - z$
 4. Then from (2), we get $x = y$

Proofs

Proof by Contraposition

- Consider a proposition $\mathcal{P} : a \Rightarrow b$
- The **converse** of $\mathcal{P}$ is $b \Rightarrow a$
- The converse of $\mathcal{P}$ need not be true
 - *If a person lives in Eugene, they live in Oregon* is true, but
If a person lives in Oregon, they live in Eugene is not.
- The **contrapositive** of $\mathcal{P}$ is $\neg b \Rightarrow \neg a$
 - A proposition and its contrapositive are **logically equivalent**:
proving $\neg b \Rightarrow \neg a$ proves $a \Rightarrow b$.

a	b	$a \Rightarrow b$	$\neg b \Rightarrow \neg a$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Proofs

Proof by Contradiction / Indirect Proof

- It is also known as *reductio ad absurdum* (reducing to absurdity).
- To prove a statement $\mathcal{Q}$, we assume $\neg\mathcal{Q}$ and derive a logical inconsistency. Since $\neg\mathcal{Q}$ cannot hold, $\mathcal{Q}$ must be true.
- To prove $\mathcal{P} : a \Rightarrow b$ by contradiction, assume **both** a and $\neg b$, and derive a contradiction.
- **Example:** $\sqrt{2}$ is irrational. Suppose $\sqrt{2} = p/q$ in lowest terms. Then $p^2 = 2q^2$, so p is even, so $p = 2k$, so $q^2 = 2k^2$ and q is even. But then p/q was not in lowest terms, a contradiction.
- Contraposition and contradiction are related but distinct: contraposition proves a different (equivalent) implication, contradiction derives an absurdity.

Proofs

Proof by Induction

- These proofs only work for propositions about integers or propositions that are indexed by integers
- To show that a statement is true for all $n \in \mathbb{N}$, we show that it is true for $n \geq n_0$ (the *base case*), and that if it is true for an arbitrary n then it is true for $n + 1$ (the *inductive step*)

Proofs

- Show that $1^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Proofs

Proof by Exhaustion

- These proofs require a finite number of cases to be possible
- Each case is proved separately
- If the proposition is true in each case, then the proposition is true

Proof by Construction

- Create a concrete example with a property to show that something having that property exists
- Create a counterexample to disprove a proposition that all elements have a certain property

Sets

- A *set* is any well-specified collection of elements.
- $x \in S$ indicates that x is an element of S .
- If $x, y \in S$, then $x = y$ denotes that x and y are the same element, and $x \neq y$ denotes that they are different elements.
- Let $\mathcal{P}$ be a property which any element $x \in S$ has or does not have.
 - $\{x \in S \mid x \text{ has property } \mathcal{P}\}$ is the set of all elements of S which have property $\mathcal{P}$.

Sets

- A set X all of whose elements belong to S is called a *subset* of S .
 - It is denoted $X \subseteq S$, read as X is *contained* in S . This **allows** $X = S$.
 - $X \subset S$ denotes a *proper* subset: $X \subseteq S$ and $X \neq S$.
 - $x \in X$ is equivalent to $\{x\} \subseteq X$
- A set which contains no elements is called the *empty set* or *null set* and is denoted by $\emptyset$.

Sets

- $A \cup B$ or A union B , is the set of all elements that are either in A or B or both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

- $A \cap B$ or A intersect B , is the set of all elements that are in both A and B .

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

Sets

- $A \setminus B$ or $A - B$ or A minus B is the set of all elements of A that are not in B .

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}$$

- $x \notin X$ expresses that x is not an element of X
- The elements of S which do not belong to X form a set called the *complement* of X in S . It is denoted by X^C .

$$X^C = S \setminus X$$

Sets

- If A and B have no element in common, i.e., $A \cap B = \emptyset$, we say they are *disjoint*.
- $A \times B$ or the *Cartesian product* of A and B is the set of pairs (a, b) where $a \in A$ and $b \in B$.
 - Let $A = \{1, 2\}$ and $B = \{x, y, z\}$
 - $A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$
- Consider m sets $S_1, S_2, \dots, S_m$. The set of m -tuples $(x_1, x_2, \dots, x_m)$, where $x_i \in S_i \forall i = 1, 2, \dots, m$ is the Cartesian product $S_1 \times S_2 \times \dots \times S_m$, also denoted $\prod_{i=1}^m S_i$.

Sets

- The *cardinality* of a set X is the number of elements of X and is denoted by $|X|$.
 - A set with cardinality 1 is called a singleton.
- A set X is *finite* if it is empty, or if there is a bijection between X and $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$.
 - $\mathbb{N}$ is the set of natural numbers.
 - Equivalently, X is finite iff $\exists n \in \mathbb{N}$ such that $|X| = n$.
- A set X is *countably infinite* if there is a bijection between X and $\mathbb{N}$.
- Finite matches an **initial segment** of $\mathbb{N}$, countably infinite matches **all** of $\mathbb{N}$.

Sets

- Sets that are either finite or countably infinite are called *countable*.
 - $\mathbb{Z}$ and $\mathbb{Q}$ are countable; $\mathbb{R}$ is not.
- The union of a finite or countable number of countable sets is countable.
- Every infinite set has a countably infinite subset.
- A *partition* of a set X refers to decomposing X into a collection of non-intersecting subsets
 - A collection of $\{X_\lambda\}_{\lambda \in \Lambda}$ of subsets of X , where Λ is a possibly infinite index set, such that

$$X = \bigcup_{\lambda \in \Lambda} X_\lambda \text{ and } \lambda_1 \neq \lambda_2 \Rightarrow X_{\lambda_1} \cap X_{\lambda_2} = \emptyset$$

Real Numbers

- $\mathbb{R}$ represents the set of all real numbers.
 - $\mathbb{R}$ is the set of all *rational* and *irrational* numbers.
- The real number line is sometimes denoted by $(-\infty, \infty)$
- Subsets of the real line specified by their end points are called *intervals*.
- For $a, b \in \mathbb{R}$ with $a \leq b$
 - The interval $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$ is an *open interval*.
 - The interval $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ is a *closed interval*.
 - The intervals $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$
or $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$ are *half-open*

Real Numbers

- Let $S \subseteq \mathbb{R}$ and let $y \in \mathbb{R}$
 - y is an *upper bound* for S if $x \leq y \ \forall x \in S$
 - y is a *lower bound* for S if $x \geq y \ \forall x \in S$.
- If y is an upper bound for S and no number smaller than y is an upper bound for S , then y is the *least upper bound* for S .
 - The least upper bound of S is the *supremum*, $\sup S$
 - If S is non-empty but not bounded above, $\sup S = +\infty$
- If y is a lower bound for S and no number larger than y is a lower bound for S , then y is the *greatest lower bound* for S .
 - The greatest lower bound is the *infimum*, $\inf S$.
 - If S is non-empty but not bounded below, $\inf S = -\infty$
- For $\emptyset$, $\sup \emptyset = -\infty$ and $\inf \emptyset = +\infty$

Real Numbers

- The supremum and infimum are unique and do not necessarily need to belong to the set S .
- If $\sup S \in S$, then it is the **maximum**
 - Consider $S = (0, 1)$
 - $x < 1, \forall x \in S$
 - $\sup S = 1$, but $1 \notin S$, so a maximum does not exist
 - Consider $S = [0, 1]$
 - $x \leq 1, \forall x \in S$
 - $\sup S = 1$ and $1 \in S$, so 1 is the maximum of S
- If $\inf S \in S$, then it is the **minimum**

Real Numbers

- **Completeness Axiom**

Every **non-empty** $S \subseteq \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$; every non-empty S bounded below has an infimum in $\mathbb{R}$.

- ▶ Completeness fails in $\mathbb{Q}$
 - $\{x \in \mathbb{Q} \mid x^2 < 2\}$ has no rational supremum.
- ▶ This axiom is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and is crucial for existence theorems.

Relations

- Let X and Y be sets. A *binary relation* $\mathcal{R}$ is simply *any* subset of $X \times Y$.
 - The domain of $\mathcal{R}$ is $\{x \in X \mid \exists y \text{ such that } (x, y) \in \mathcal{R}\}$
 - The range of $\mathcal{R}$ is $\{y \in Y \mid \exists x \text{ such that } (x, y) \in \mathcal{R}\}$
- $x\mathcal{R}y$ denotes that x is related to y .
- $\mathcal{R}$ is an *equivalence relation* on S if it satisfies:
 1. *reflexivity*, $x\mathcal{R}x \ \forall x \in S$
 2. *symmetry*, $x\mathcal{R}y \Rightarrow y\mathcal{R}x \ \forall x, y \in S$
 3. *transitivity*, $x\mathcal{R}y \wedge y\mathcal{R}z \Rightarrow x\mathcal{R}z \ \forall x, y, z \in S$.
- An equivalence relation on S induces a partition of S into *equivalence classes*.

Relations

- $\mathcal{R}$ is called a *partial ordering* $\succsim$ on S if it satisfies:
 1. reflexivity
 2. transitivity, and
 3. *antisymmetry*, $x\mathcal{R}y \wedge y\mathcal{R}x \Rightarrow x = y \quad \forall x, y \in S$
 - The set S is then called a *partially ordered set* or a *poset*.
- If in addition any two elements are comparable ($x \succsim y$ or $y \succsim x \quad \forall x, y \in S$), the ordering is *complete* or *total*.
 - Completeness and transitivity of $\succsim$ are the standard axioms on preferences.

Functions

- Let X and Y be two sets.
- If each element $x \in X$ is associated with one and only one element $y \in Y$, a *function* f from X into Y is defined.

$$f : X \rightarrow Y$$

- f is also called a *transformation*, *map* or *operation* from X into Y .
- x is the variable, y is the image of x by f , or the value of f at x .

$$y = f(x)$$

Functions

- The *domain* of a function f is the set of all x such that $f(x)$ is defined.
- The *range* (or image) of f is $f(X) = \{f(x) \mid x \in X\} \subseteq Y$.
- Y itself is the *codomain*. The range may be a proper subset of the codomain.

Functions

- If all elements of X have the same image in Y , we say that f is constant.
- Let $A \subseteq Y$. The *inverse image* (or preimage) of A under f is

$$f^{-1}(A) = \{x \in X \mid f(x) \in A\}$$

- ▶ $f^{-1}(A)$ is defined for every f ; it does not require f to be invertible.

Functions

- f is *injective* (one-to-one) if $\forall y \in Y$ there is *at most* one $x \in X$ such that $f(x) = y$.
 - Equivalently, $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.
- f is *surjective* (or *onto*) if $\forall y \in Y, \exists$ *at least* one $x \in X$ such that $f(x) = y$.
 - Otherwise f maps X *into* Y .
- f is *bijective* if $\forall y \in Y, \exists! x \in X$ such that $f(x) = y$.
 - f is bijective iff it is both injective and surjective.
 - f has an inverse function $f^{-1} : Y \rightarrow X$ **iff** f is bijective.